

Auteur: Odia Kibor
 Studierichting: Informatica
 Oefeningenreeks 1D nr. 21.16

$$z^5 + 7z^4 + 4z + 28$$

-7 is een nulpunt

$$\begin{array}{c|cccccc} -7 & 1 & 7 & 0 & 0 & 4 & 28 \\ & & -7 & 0 & 0 & 0 & -28 \\ \hline & 1 & 0 & 0 & 0 & 4 & 0 \end{array}$$

$$\Rightarrow (z + 7)(z^4 + 4)$$

Ontbindt $z^4 + 4$

$$z^4 + 4 \rightarrow z^4 = -4 \rightarrow z^2 = \pm 2i$$

Voor $z^2 = 2i$

$$\Rightarrow (x + yi)^2 = x^2 - y^2 + 2xyi$$

$$\begin{cases} x^2 - y^2 = 0 \\ 2xy = 2 \end{cases} \Rightarrow \begin{cases} x^2 - y^2 = 0 \\ xy = 1 \\ (xy < 0) \end{cases} = \begin{cases} 0 \\ 1 \end{cases} \Rightarrow \begin{cases} x^2 - y^2 = 0 \\ x^2(-y^2) = -1 \end{cases}$$

$$\lambda^2 - 1 = 0 \Rightarrow \lambda^2 = 1 \Rightarrow \lambda = \pm 1$$

$$x^2 = 1 \Rightarrow x = \pm 1$$

$$-y^2 = -1 \Rightarrow y = \pm 1$$

Oplossing voor $2i$: $1 + i$ of $-1 - i$

Voor $z^2 = -2i$

$$\Rightarrow (x + yi)^2 = x^2 - y^2 + 2xyi$$

$$\begin{cases} x^2 - y^2 = 0 \\ 2xy = -2 \end{cases} \Rightarrow \begin{cases} x^2 - y^2 = 0 \\ xy = -1 \\ (xy > 0) \end{cases} = \begin{cases} 0 \\ -1 \end{cases} \Rightarrow \begin{cases} x^2 - y^2 = 0 \\ x^2(-y^2) = -1 \end{cases}$$

$$\lambda^2 - 1 = 0 \Rightarrow \lambda^2 = 1 \Rightarrow \lambda = \pm 1$$

$$x^2 = 1 \Rightarrow x = \pm 1$$

$$-y^2 = -1 \Rightarrow y = \pm 1$$

Oplossing voor $-2i$: $1 - i$ of $-1 + i$

nulpunten: $-7, (1 + i), (-1 - i), (1 - i), (-1 + i)$

References